

Roll No.....

MID SEMESTER EXAMINATION March - 2025

Name of the Program: B. Tech (EE)

Semester: 8th

Name of the Course: Expert System & Fuzzy Logic

Course Code: TOE 801

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	What are the different Components of an Expert System? Explain with Suitable diagrams.	
OR		
(b)	How do computing techniques enhance Expert System functionalities? Explain with suitable examples.	CO1
Q2	(10 marks)	
(a)	How do forward chaining and backward chaining work in Expert Systems? Explain with suitable diagrams.	
OR		CO1
(b)	Explain the concept of Rule-Based Systems and their applications in Expert Systems.	
Q3	(10 marks)	CO2
(a)	Why are Expert systems important in the present time? Explain with Examples and Figures.	
OR		
(b)	Discuss some limitations and challenges faced in Expert System development.	CO2
Q4	(10 marks)	
(a)	Discuss the classifications of expert systems in artificial intelligence.	
OR		CO2
(b)	What is the role of inference engines in Expert Systems? Explain with suitable diagrams.	
Q5	(10 marks)	CO2
(a)	What are the practical uses of Expert Systems across different industries?	
OR		
(b)	What are DENDRAL and DCplain? How do they assist in building Expert Systems?	CO2

